

RPS METOCEAN DATAFEEDS SECURED WEB SERVICES.

RPS offers clients and their representatives' access to data collected on their behalf, from the RPS DataFeeds system.

DataFeeds has a RESTful interface allowing clients to access, request and ingest environmental data, in formats suitable for consumption further downstream.

The RESTful interface is a modern, standards (TLS/SSL) protected, API that requires end-users to be in possession of an RPS supplied certificate. The certificate provides the means to authenticate encrypted sessions with DataFeeds, within which URLs can be supplied to query for, and extract relevant data. In cases where data-push models are better suited, the interface can be used for data ingress.

Datasets available are aligned with an authorisation matrix, limiting access to data associated with the username(s) embedded within the supplied certificate(s).

Clients accessing the DataFeeds RESTful services must make the request using the Secure Socket Layer (SSL) certificate signed by the MetOcean Certificate Authority (CA). Clients will need to install the certificate obtained from RPS MetOcean.

Data is transferred within the body of the HTTP request/response as formatted JSON (Java-Script Object Notation) text by default. CSV (Comma-Separated Values) and KML formats are also available.

Secured URL:

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured>

API Specification

GET /list[.json]

Get a list of data available.

The return object contains a summary of all feed streams available. This is returned in either a JSON or free text format depending on your API request. This is where you obtain the **feedID** required to access the data.

Endpoint URL

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/list.json>

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/list>

Example Response

Your **feedID** is the “Stream/feedID” property value in the free text response. Shown in the example below, your {**feedID**} is “A14WaveFcst-fs”.

```
-----  
Feed           : A14 Forecast Wave  
Stream/feedID  : A14WaveFcst-fs  
Enabled        : true  
Client         : RPS Australia West Pty Ltd  
Project        : A0550.001  
Location       : A14  
-----
```

Your **feedID** is the “feedID” property value in the JSON response. Shown in the example below, your **{feedID}** is “A14WaveFcst-fs”.

```
streamSummaryList :[
{
  details : "services/feeds/A14WaveFcst-fs",
  "client": "RPS Australia West Pty Ltd",
  project": "A0550.001",
  "feedID": "A14WaveFcst-fs",
  enabled": true,
  location": "A14",
  displayName": "A14 Forecast Wave"
}
]
```

GET /timeseries.v2/{feedID}.json|csv

Get data for all or a sub-set of variables for a specified feedID within a time period.

The returned object contains the dimensionality of the data, global attributes, a list of variables and their associated attributes and dimensional data records. Being a secured service, this will also verify that the request has a certificate and that the user (specified in the certificate) is allowed access to the particular feed-stream.

Endpoint URL

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/timeseries.v2/{feedID}.json>

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/timeseries.v2/{feedID}.csv>

Query Parameters

Name	Type	Required	Description
from	date	required	YYYYMMDDHHMM specifies the start date time of the data to retrieve. YYYY - Year MM - Month (01 = January) DD - Day (01 = First day of month) HHMM – Hour Minutes (0 – 24 hours; 0 – 59 minutes) <i>For example, 201606011200 request data from 12:00 1st Jun 2016.</i>
to	date	required	YYYYMMDDHHMM specifies the last date time of the data to retrieve. If not specified, data will be retrieve to the latest available.
tz	integer	optional	The timezone of the from/to parameter. If not specified, default is the acquisition timezone.
variables	string	optional	Specifies a list of variables to be retrieved; comma separated. If not specified, all available variables will be retrieved. <i>For example, variables=WindSpd,WindDirn or WindSpd[1-3,5],WindDirn[1-3,5]</i>
excludeVariables	string	optional	Specifies a list of variables to exclude; comma separated. By default, all variables (including multi-dimensional variables) are returned.
dataFilter	boolean	optional	If set to true returns a subset of the recommended variables. This filter is most useful in the case of where multiple variables of the same data type exist in the feed.
subSample	integer	optional	Specifies the sub sample interval in <i>seconds</i> for time series data to be retrieved. This is RECOMMENDED for high frequency data.

subSampleTolerance	integer	optional	Specifies the tolerance of the sub sample interval in <i>seconds</i> for time series data to be retrieved. The closest match to the sub sample time will be returned.
subSampleOffset	integer	optional	Specifies the sub sample offset from 0000 hours in seconds for time series data to be retrieved.
mem	boolean	optional	If set to true, returns the object with MEM (Maximum Entropy Method) wave spectra.
cos2s	boolean	optional	If set to true, returns the object with parameterised wave spectra. <i>Default is true.</i>
csvHeader	boolean	optional (<i>csv endpoint</i>)	If set to true returns header information at the top of the CSV (text) output. <i>Default is true.</i>
csvColumnHeader	boolean	optional (<i>csv endpoint</i>)	If set to true returns column header information for each data column in the CSV (text) output. <i>Default is true.</i>
noMultiDimVars	boolean	optional (<i>csv endpoint</i>)	If set to true returns only single dimensional timeseries data in the CSV (text) output.
precision	integer	optional (<i>csv endpoint</i>)	Specifies precision of CSV (text) output.
scientific	boolean	optional (<i>csv endpoint</i>)	If set to true returns scientific notation for CSV (text) output.

GET /latest.v2/{feedID}.json|csv

Get the latest available time and values for specified feedID.

The returned object is of the same form as the time-series request, but with only 1 record. Being a secured service, this will also verify that the request has a certificate and that the user (specified in the certificate) is allowed access to the particular feed-stream.

Endpoint URL

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/latest.v2/{feedID}.json>

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/latest.v2/{feedID}.csv>

Query Parameters

Name	Type	Required	Description
tz	integer	optional	The timezone of the from/to parameter. If not specified, default is the acquisition timezone.
variables	string	optional	Specifies a list of variables to be retrieved; comma separated. If not specified, all available variables will be retrieved. <i>For example, variables=WindSpd,WindDirn or WindSpd[1-3,5],WindDirn[1-3,5]</i>
excludeVariables	string	optional	Specifies a list of variables to exclude; comma separated. By default, all variables (including multi-dimensional variables) are returned.
dataFilter	boolean	optional	If set to true returns a subset of the recommended variables. This filter is most useful in the case of where multiple variables of the same data type exist in the feed.

mem	boolean	optional	If set to true, returns the object with MEM (Maximum Entropy Method) wave spectra. <i>Default is true.</i>
cos2s	boolean	optional	If set to true, returns the object with parameterised wave spectra. <i>Default is true.</i>
csvHeader	boolean	optional (csv endpoint)	If set to true returns header information at the top of the CSV (text) output. <i>Default is true.</i>
csvColumnHeader	boolean	optional (csv endpoint)	If set to true returns column header information for each data column in the CSV (text) output. <i>Default is true.</i>
noMultiDimVars	boolean	optional (csv endpoint)	If set to true returns only single dimensional timeseries data in the CSV (text) output.
precision	integer	optional (csv endpoint)	Specifies precision of CSV (text) output.
scientific	boolean	optional (csv endpoint)	If set to true returns scientific notation for CSV (text) output.

GET /timeseries.v2/{feedID}.kml

Get a KML file format for a specified feedID within a time period.

The returned object is a geographic data set for display on Earth browsers such as Google Earth.

Endpoint URL

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/timeseries.v2/{feedID}.kml>

Query Parameters

Name	Type	Required	Description
from	date	required	YYYYMMDDHHMM specifies the start date time of the data to retrieve. YYYY - Year MM - Month (01 = January) DD - Day (01 = First day of month) HHMM – Hour Minutes (0 – 24 hours; 0 – 59 minutes) <i>For example, 201606011200 request data from 12:00 1st Jun 2016.</i>
to	date	required	YYYYMMDDHHMM specifies the last date time of the data to retrieve. If not specified, data will be retrieve to the latest available.
tz	integer	optional	The timezone of the from/to parameter. If not specified, default is the acquisition timezone.
variables	string	optional	Specifies a list of variables to be retrieved; comma separated. By default, only time and geographic position data are returned. See JSON response for available variables. <i>For example, variables=Hs,HsSwell</i>

subSample	integer	optional	Specifies the sub sample interval in <i>seconds</i> for time series data to be retrieved. This is RECOMMENDED for high frequency data.
subSampleTolerance	integer	optional	Specifies the tolerance of the sub sample interval in <i>seconds</i> for time series data to be retrieved. The closest match to the sub sample time will be returned.
subSampleOffset	integer	optional	Specifies the sub sample offset from 0000 hours in seconds for time series data to be retrieved.

GET /latest.v2/{feedID}.kml

Get a KML file format for a specified feedID for the latest available geographic position.

The returned object is a geographic data set for display on Earth browsers such as Google Earth.

Endpoint URL

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/latest.v2/{feedID}.kml>

Query Parameters

Name	Type	Required	Description
tz	integer	optional	The timezone of the from/to parameter. If not specified, default is the acquisition timezone.
variables	string	optional	Specifies a list of variables to be retrieved; comma separated. By default, only time and geographic position data are returned. See JSON response for available variables. <i>For example, variables=Hs,HsSwell</i>

Example Request and Responses

Response Codes:

- **200** — Success. Dataset added to the queue of data to ingest.
- **400** — Failure. Invalid request due to an invalid format, or limit exceeded.
- **401** — Failure. Unauthorised request.
- **404** — Failure. feedID or datasetUID not recognized.
- **500** — Failure. Internal error occurred.

Example JSON Request

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/timeseries.v2/A14WaveFcst-fs.json?from=20160101000&tz=8>

Returns the whole dataset in JSON text from the period of 10:00 1st Jan 2016 WST (UTC+08:00) to present.

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/timeseries.v2/A14WaveFcst-fs.json?from=20160101000&tz=8&dataFilter=true>

Returns the “recommended” dataset in JSON text from the period of 10:00 1st Jan 2016 WST (UTC+08:00) to present. For example, a feed with multiple anemometers the “recommended” dataset may contain just the 10 minute mean winds from the best sited anemometer

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/timeseries.v2/A14WaveFcst-fs.json?from=20160101000&tz=8&variables=Time:Latitude:Longitude:BP:AirTemp>

Returns a subset of the dataset in JSON text from the period of 10:00 1st Jan 2016 WST (UTC+08:00) to present.

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/latest.v2/A14Met-mqtt.json>

Returns the latest time record available for the whole dataset in JSON text; only 1 time record will be returned.

Example JSON data format response:

```
{
  "dimensions": {
    "records": 0,
  },
  "attributes": {
    "project": {"type": "string", "value": "J3068"},
    "expectedRecordInterval": {"type": "numeric", "value": 600},
    "client": {"type": "string", "value": "RPS Evans-Hamilton, Inc"},
    "location": {"type": "string", "value": "Liza 1 Mooring C"}, ....
  },
  "constants": {
    "DepthHeight": {
      "primitiveType": "numeric",
      "dataType": "DepthHeight",
      "dimensionNames": [],
      . . . .
      "data": 26.5,
      "quality": null
    },
    . . . .
  },
  "variables": {
    "Time": {
      "primitiveType": "datetime",
      "dataType": "Time",
      "dimensionNames": ["records"],
      "attributes": {
        "timezone": {"type": "numeric", "value": 0},
        "description": {"type": "string", "value": "milliseconds since 1970-01-01T00:00:00+0000"},
        "longName": {"type": "string", "value": "Time"},
        "units": {"type": "string", "value": "ms"}},
        . . . .
      },
    "BP": {
      "primitiveType": "numeric",
      "dataType": "BP",
      "dimensionNames": ["records"],
      "attributes": {
        "missingValue": {"type": "numeric", "value": -1.9938419936773738e+37},
        "longName": {"type": "string", "value": "Barometric Pressure"},
        "fillValue": {"type": "numeric", "value": -1.9938419936773738e+37},
        "units": {"type": "string", "value": "hPa"}},
        . . . .
      },
    . . . .
  },
  "data": [
    {
      "time": "2017-02-08T04:00:00.000Z",
      "variables": [1486526400000, 1012.7328491210938],
      "quality": [null, null]
    },
    {
      "time": "2017-02-08T04:10:00.000Z",
      "variables": [1486527000000, 1012.6253051757812],
      "quality": [null, null]
    },
    . . .
  ]
}
```

Example CSV Request

<https://datafeeds.rpsmetocean.com:8443/datafeeds-web/services/feeds/secured/timeseries.v2/A14Current-rudics.csv?from=20160101000&tz=8&dataFilter=true>

Returns the “recommended” dataset in CSV text from the period of 10:00 1st Jan 2016 WST (UTC+08:00) to present.

Example CSV data format response:

```
:client = Your Company Pty Ltd
:project = A0420
:location = A14
:logging_system = datawell
:transport_provider = iridium
:arrival_mechanism = http
:data_type = current
:data_origin = measured
:data_dimension = point
:expected_record_interval = 600.0
```

```
Time,BatteryLife (s),BatteryVoltage (volt),CurDirn (degrees),CurSpd (m s-1),Height (m),Longitude (degrees_E),SeaWaterTemp (degree_C)
```

```
2017-11-30T16:00:00.000+0800,41.000000,20.600000,332.747253,0.476000,0.000000,123.277917,30.890000
2017-11-30T16:10:00.000+0800,41.000000,20.600000,328.527473,0.446000,0.000000,123.277831,30.890000
2017-11-30T16:20:00.000+0800,41.000000,20.500000,329.142857,0.473000,0.000000,123.277767,30.890000
2017-11-30T16:30:00.000+0800,41.000000,20.500000,326.065934,0.469000,0.000000,123.277724,30.890000
2017-11-30T16:40:00.000+0800,41.000000,20.500000,321.582418,0.384000,0.000000,123.277703,30.890000
2017-11-30T16:50:00.000+0800,41.000000,20.500000,319.296703,0.367000,0.000000,123.277660,30.990000
2017-11-30T17:00:00.000+0800,41.000000,20.500000,318.857143,0.387000,0.000000,123.277617,30.990000
```

Additional Information on SSL Certificate

The SSL certificate obtained from RPS is self-signed by the Metocean Certificate Authority (CA) in which you will get a “CA Root Certificate is not trusted because it is not in the Trusted Root Certification Authorities store” message. If you want to allow access to the server you will need to disable the verify peer. The connection retains as an authenticate encrypted session.

Converting from PKCS12 (.p12) to PEM

```
openssl pkcs12 -in {SSL_CERTIFICATE_NAME}.p12 -out {SSL_CERTIFICATE_NAME}.pem -clcerts  
openssl pkcs12 -in {SSL_CERTIFICATE_NAME}.p12 -out {SSL_CERTIFICATE_NAME}.key -  
nocerts -nodes ( enter passphrase )
```

Note: If using openssl v3, you will need to use the `-legacy` option

Accessing using CURL

```
curl -k https://datafeeds.rpsmetocean.com:8443/datafeeds-  
web/services/feeds/secured/latest.v2/{feedID}.json --key {SSL_CERTIFICATE_NAME}.key -  
-cert {SSL_CERTIFICATE_NAME}.pem -o {feedID}.json --silent --verbose
```

Accessing via Python

```
# turn off warnings  
urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)  
response = requests.get(URL,  
cert=[{SSL_CERTIFICATE_NAME}.pem, {SSL_CERTIFICATE_NAME}.key], verify=False)
```